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Cryptology Assignment 3

July–November 2025

4-Round DES Differential Cryptanalysis

Instructions

- This assignment focuses on breaking a **custom 4-round DES variant** using differential cryptanalysis. Your goal is to recover the 56-bit secret key assigned to your oracle.
- Queries can be sent through the **web interface** and the **programmatic API**. Refer to `query.ipynb` for detailed usage examples.
- **Documentation matters**: Include a clear `README.md` describing how to run your code.
- **Academic Integrity**: This is an individual assignment. Plagiarism, if detected, will result in serious consequences, as discussed in class.

Problem Statement

You are provided with a Python implementation of a custom 4-round DES-like cipher, complete with modified S-boxes. An encryption oracle is available for your roll number. Your task is to:

- Analyze the cryptographic properties of the provided S-boxes
- Design and execute a differential cryptanalysis attack
- Recover the complete 56-bit secret key for your assigned instance

Oracle Access

Web Interface

`https://des-oracle.bombaykettle.org/<your-roll-number>`

Example: `https://des-oracle.bombaykettle.org/cs22btech11004`

Programmatic API

Endpoint: https://des-oracle.bombaykettle.org/<your-roll-number>/api/encrypt

See ***query.ipynb*** *for implementation details and the format of the request.*

Provided Materials

- `custom_DES.py` – Implementation of the 4-round DES variant
- `S_BOXES.txt` – Modified S-box definitions
- `query.ipynb` – API usage examples and helper code

Tasks

1. **S-Box Analysis**
   - Compute the **Differential Distribution Table (DDT)** for each S-box
   - Identify high-probability input/output differentials suitable for attack
   - Document the most promising differential characteristics
2. **Differential Pair Construction**
   - Design plaintext pairs with carefully chosen XOR differences
   - Query the encryption oracle with your constructed pairs
   - Collect and organize the corresponding ciphertext pairs
3. **Key Recovery**
   - Extract subkey bits from the final round using differential analysis
   - Progressively recover subkeys from earlier rounds
   - Reconstruct the complete 56-bit master key
   - **Verify** your solution by decrypting test ciphertexts

Deliverables

Submit a **single ZIP file** named `<RollNo>_Assgn3.zip` (e.g., `cs22btech11004_Assgn3.zip`) containing:

1. `key.txt`
   - Contains **only** the recovered 14-character hexadecimal key
   - Format: `0123456789ABCD`

2. `report.pdf`
    - Content:
        - DDT analysis methodology and key findings
        - Complete differential trail with probability calculations
        - Step-by-step key recovery process
3. `code/`
    - All source files (`.py` and/or `.ipynb`)
    - Scripts for DDT generation, differential analysis, and key recovery
    - Well-commented code
4. `README.md`
    - Steps to run the code and output the key.

Good luck with your cryptanalysis!